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Original Research

Does economic policy uncertainty drive foreign direct investment inflows? Evidence from South Africa

Babatunde Lawrence[a], Fabian Moodley[b,*], Sune Ferreira-Schenk[b]
[a]Trade Research Entity, North-West University, Potchefstroom, South Africa
[b]School of Economic Science, North-West University, Potchefstroom, South Africa

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ABSTRACT

Foreign direct investment (FDI) is central to economic growth in emerging economies, yet the extent to which economic policy uncertainty (EPU) shapes FDI inflows in African contexts remains empirically underexplored, particularly using dynamic, direction-sensitive methods. This study examines the time-varying and directional relationship between EPU and FDI in South Africa using monthly data spanning January 2006 to December 2021, with FDI sourced from the World Bank and EPU proxied by the World Uncertainty Index. A time-varying parameter vector autoregression (TVP-VAR) connectedness framework is applied, complemented by the Maximal Overlap Discrete Wavelet Transformation (MODWT) across four time scales to test robustness over different investment horizons. Results show that overall connectedness between EPU and FDI is low (TCI = 1.15), yet a clear asymmetry emerges: EPU acts as a net transmitter of shocks while FDI functions as a net receiver, with spillovers intensifying during periods of instability. MODWT confirms that this negative influence strengthens over medium- to long-term horizons. By combining dynamic connectedness and multi-scale wavelet evidence, the study fills a gap in emerging-market literature and offers implications for fostering policy stability to sustain foreign investment inflows.

* Corresponding author.
E-mail address: fabian.moodley@nwu.ac.za
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1. Introduction

Foreign direct investment (FDI) is critical to economic growth, technology transfer, employment generation, and financial development, particularly in developing economies. FDI as a form of external capital that supplements domestic savings and contributes to long-term development objectives is a significant issue in the case of South Africa. But the sustainability and stability of the FDI inflow to the country are largely affected by macroeconomic conditions, institutional quality and, most importantly, economic policy uncertainty (EPU). The rapid rate of changes in governments, new policies and shocks to the global economy in recent years has increased uncertainty leading some investors to question the impact of such uncertainty on investment decisions. EPU has become a leading factor for the investment behavior of firms due to firms' expectations of future earnings, economic regulation, and macroeconomic stability. Based on the real options theory, a degree of uncertainty increases the value of wait-time, prompting investors to postpone or decrease irreversible investments, like FDI. This is especially important in developing countries, where structural fragility and policy incoherence could amplify uncertainty shocks. South Africa makes a salient case here because it has experienced domestic policy oscillations and global financial conditions throughout the last 20 years.

Although EPU has an increasingly considerable impact on investment flows, the empirical evidence on whether EPU has a significant effect on FDI is mixed and remains relatively unexplored from an African country perspective. Some find a negative relationship between uncertainty and investment, while others report nonlinearities, asymmetries, and time-varying effects that static models cannot account for. Furthermore, in the literature, most studies are based on classical econometric methods that overlook the dynamic and directional nature of the correlation between EPU and FDI. Therefore, this research has limited insight into the transmission of shocks to policy uncertainty to foreign investment and whether this transmission is symmetric or evolves across different economic regimes.

This study fills this gap by using a time-varying parameter vector autoregression (TVP-VAR) approach to investigate the dynamic links between EPU and foreign direct investment in South Africa. While traditional models can only model a single factor of the network, the TVP-VAR model shows that spillovers are dynamic and therefore enable the discovery of directional relationships of variables. The study provides a more nuanced analysis of the transmission mechanism by constructing generalized forecast error variance decompositions and connectedness indices that measure the manner in which EPU shocks become transferable to FDI and vice versa.

This observation indicates that the overall connectedness between EPU and FDI is low, implying that both variables are mostly driven by their own innovations. But directionality analysis reveals a stark asymmetry: EPU acts as a net transmitter of shocks and foreign direct investment acts as a net receiver. This means that policy uncertainty largely affects investment inflows, but FDI fluctuations do not change the level of uncertainty. In addition, it is further established through dynamic analysis that the variation of it is due to the strength and time dependence. The strength of this relationship is likely to become stronger with the instability of economic and political instability.

By focusing on South Africa, this study contributes to the literature in several important ways. First, it provides new evidence on the uncertainty–investment nexus in an emerging market context that has received relatively limited attention. Second, it advances the methodological approach by applying a dynamic connectedness framework to capture time-varying and directional spillovers between EPU and FDI. Third, the findings offer important policy implications, highlighting the need for stable and predictable policy environments to attract and sustain foreign investment.

The remainder of the paper is structured as follows. Section 2 reviews the relevant theoretical and empirical literature. Section 3 outlines the data and methodology, including the TVP-VAR connectedness framework. Section 4 presents and discusses the empirical results. Finally, Section 5 concludes with policy implications and suggestions for future research.

2. Literature Review

In recent years, in the light of rising global economic instability, the nexus between the factors of EPU and FDI has gained increasing relevance. FDI is well accepted to drive economic growth, technical improvement, and employment growth and this is particularly true in emerging economies such as South Africa. The extent of this impact, however, is extremely sensitive to macroeconomic settings and the nature of the institutions created and the policies enacted within those environments. EPU has become one of such drivers affecting investment decisions. The empirical study results of the research are mixed regarding the influence of EPU on FDI. A large number of studies indicate that greater riskier investment is disincentivized from investment by increasing risk and decreasing expected yield (Bloom, 2009; Gulen and Ion, 2016).

Under such circumstances, which are known as uncertainty, firms would delay (or even scale back) investment decisions, especially if the investment is irreversible. Studies such as Canh et al. (2020) and Phan et al. (2021) suggest that EPU adversely influences investment (including FDI) flows due to the increase in volatility and reduced investor confidence. In contrast, some research points out that uncertainty and investment are not strictly linear or symmetrically related. Bahmani-Oskooee and Saha (2019) note that in specific contexts uncertainty has an effect depending on the relative size and level of persistence and the economic context. In some moderate instances, it is plausible that uncertainty has no serious impacts on investment and severe uncertainty has its deleterious effects. Ramakau et al. (2025b) examined the dynamics of equity market volatility and EPU in South Africa during the COVID-19 pandemic. The study captured volatility clustering, persistence, and asymmetry across sectors. The results indicated that volatility was highly persistent during the pandemic, and different economic sectors reacted differently to shocks. In addition, the effects of EPU may differ between countries, particularly developed and emerging countries given varying institutional quality and market structures (Fedderke and Romm, 2006; Kariuki, 2015). Both emerging markets' external shocks and domestic structural constraints frequently exacerbate the impact of uncertainty. Parsons and Krugell (2022) argue that uncertainty clearly has a strong impact on economic activity in South Africa and suggest that investment decisions are very sensitive to policy instability. Similarly, Busse and Hefeker (2007) and Henisz (2000) argue that political and institutional risks are very important in shaping FDI inflows since investors will seek stable and predictable policy regimes. Ramakau et al. (2025a) analysed stock market volatility and EPU of South Africa (SA), the United States of America (USA), Brazil, Russia, India, and China (BRIC) during the COVID-19 pandemic. The study found SA and China to be the most volatile during the COVID-19 pandemic where also spillover effects from the USA to the emerging countries were found to be small but statistically significant. SA showed the strongest impact from USA market shocks. Brazil's and India's equity markets were found to be highly sensitive to changes in EPU relative to the other countries.

Nevertheless, previous literature has mostly employed classical econometric models, which may not provide adequate predictions of dynamic and time-dependent relationships between EPU and FDI. In response to these shortcomings, dynamic modelling methods have also been used in numerous recent studies. For example, Diebold and Yilmaz (2012) stressed the importance of investigating spillover effects and economic variables' connectivity. Such methods show that the relationship between uncertainty and investment is not static but evolves over time, particularly during periods of economic instability or political upheaval. However,

little is known about the directionality of the transmission of shocks between EPU and FDI, especially in African countries. This study contributes to the literature by applying a time-varying connectedness framework to better understand this dynamic within the South African context.

2.1. Theoretical Review

The theoretical nexus between EPU and foreign direct investment is based on economic theories such as real options theory, neoclassical investment theory, institutional theory, and uncertainty-based macroeconomic theory. The real options theory offers a key theoretical perspective in determining the way that uncertainty influences the investment decisions. Investment is assumed by Dixit and Pindyck (1994) to be irreversible and entails high sunk costs. Because of this uncertainty, firms may choose to delay investments in order to gather more information about future economic conditions. That "wait-and-see" behavior has the effect of accelerating the value of procrastinating investment with the most uncertainty being in the economy.

This argument is also supported by Bloom (2009) who posits that periods of uncertainty shocks lead to a loss of investment and economic activity as a result of a brief state of uncertainty as firms are more guarded with their capital decisions. In the case of FDI, it appears that foreign firms will delay or withdraw investment in countries with high levels of policy uncertainty. According to neoclassical investment theory (Jorgenson, 1963), investment decisions are influenced by expected return, cost of capital and economic fundamentals. This model presupposes rational decision-making under certainty but can be generalized to include uncertainty as it acknowledges that rising policy uncertainty increases the cost of ownership of capital and reduces the anticipated expected return.

In consequence, increased EPU can discourage investment by raising the risk premium and dampening the appeal of the host country to overseas investors. Institutional theories are similarly important to grasp the relationship between EPU and FDI. North (1990) insists that institutions are the formal rules and regulations that govern economic activity and influence the behavior of economic actors. Stable and transparent institutions reduce uncertainty and transaction costs, making it easier to attract FDI. FDI then becomes easier to finance. Conversely, when policy frameworks are weak or inconsistent, the effect is to create uncertainty and dissuade FDI even more. Indeed, Henisz (2000) puts it another way: political constraints and institutional quality are crucial determinants of investment decisions, as they govern the reliability and effectiveness of government policies. Uncertainty-based macroeconomic theories emphasize the wider economic and performance impacts of policy uncertainty. Bloom (2014) and Karnizova and Li (2014) argue that shocks to uncertainty can lead to a variation in economic activities due to their impact on consumption, investment, and financial markets. These types of shocks travel through the economy and generate volatility and reduce economic performance as a whole. In emerging markets where economic structures are often tenuous, the effect of uncertainty is expected to be much greater.

Furthermore, regime-switching and time-variable models (Ang and Timmermann, 2012; Hamilton, 1989) support the idea that the relationship between uncertainty and investment may vary according to economic conditions. In times of stability, uncertainty might have only a slight impact, but during periods of crisis or political upheaval, its effects could be magnified. This favors the adoption of dynamic models able to represent changing relationships over time. These theoretical perspectives, when considered together, indicate that foreign direct investment is adversely influenced by EPU through the extension of policy risk, the postponement of policy investment decisions, and erosion of investor confidence. Nevertheless, the nature and magnitude of this association may depend on the quality of

institutional and financial systems, economic circumstances and the fluid pattern of uncertainty shocks.

3. Methodology

The study implements monthly time series of FDI and EPU for the period January 2006 to December 2021 to examine the impact of EPU on foreign direct investment under TVP-VAR conditions in South Africa. The data frequency and sample period were dictated by the availability of data as only monthly data were available for these variables. FDI data were extracted from the World Bank database, whereas EPU was proxied by the World Uncertainty Index (WUI). The WUI is widely used in the recent literature such as Msomi et al. (2026). The WUI was constructed to ensure comparability across time periods and countries. The WUI data were obtained from its central repository[1].

3.1. TVP-VAR Model

This study employs the TVP-VAR model of Antonakakis et al. (2020) to establish connectedness and the EPU transmission effect on FDI. The TVP-VAR model with one lag can be expressed as follows:

$$\Delta y_t = \theta_t \Delta y_{t-1} + \mu_t \text{ where } \mu \sim N(0, E_t) \quad (1)$$

$$vec(\theta) = vec(\theta_{t-1}) + r_t \;\; r_t \sim N(0, Q_t) \quad (2)$$

where Δy_t, Δy_{t-1} and μ_t are vectors of $N \times 1$ dimension, θ_t and E_t are matrices of $N \times N$ dimension, $vec(\theta)$ and r_t are parameter matrices of $N^2 \times 1$ dimension, Q_t is an $N^2 \times N^2$ dimensional matrix for sectoral index returns. In line with Antonakakis et al. (2020), the TVP-VAR is estimated and then transformed into a time-varying parameter vector moving average (TVP-VMA) representation using the Wold representation theorem (WRT). WRT is defined as:

$$\Delta x_t = \sum_{i-1}^{p} \beta_{it} \Delta x_{t-1} + \varepsilon_t = \sum_{j=1}^{\infty} \gamma_{jt} \epsilon_{t-j} + \varepsilon_t. \quad (3)$$

Subsequently, the TVP-VMA coefficients are mined to calculate the generalized forecast error variance decomposition (GFEVD) developed by Koop et al. (1996) and Pesaran and Shin (1998), from which the Diebold and Yilmaz (2012) connectedness index is constructed. The pairwise directional connectedness from *j* to *i* which portrays the impact of sector *j* on sector *i* in terms of its forecast error variance, is derived from the unscaled GFEVD, defined as:

$$\emptyset_{ij,t}^{g}(J) = \frac{\Sigma_{ii,t}^{-1} \Sigma_{t=1}^{J-1} \left(l_j' A_t E_t l_j\right)^2}{\Sigma_{j=1}^{N} \Sigma_{t=1}^{J-1} \left(l_i A_t \Sigma_t A_t' l_i\right)} \quad (4)$$

$$\varphi_{ij,t}^{g}(J) = \frac{\emptyset_{ij,t}^{g}(J)}{\Sigma_{j=1}^{N} \emptyset_{ij,t}^{g}(J)} \quad (5)$$

where $\Sigma_{j=1}^{N} \varphi_{ij,t}^{g}(J) = 1$ and $\Sigma_{i,j=1}^{N} \varphi_{ij,t}^{g}(J) = N$, where *J* represents the forecast horizon and l_i is a selection vector that is equal to 1 at the ith position and 0 otherwise. Using the GFEVD, the total connectedness index is constructed as given by:

[1] available at: https://worlduncertaintyindex.com (accessed in July 2025).

$$C_t^g(J) = 1 - N^{-1} \sum_{i-1}^{N} \varphi_{ii,t}^g \ (J) \quad (6)$$

This connectedness approach demonstrates how a shock in EPU can spill over into FDI in South Africa. Assume that country *i* transmits its shock to the remaining country *j*, and the resulting total directional connectedness to others is given by:

$$C_{i\to j,t}^g(J) = \sum_{j=1,i\neq j}^{N} \varphi_{ji,t}^g(J) \quad (7)$$

Furthermore, the directional connectedness country *i* receives from country *j* is termed total directional connectedness from others and is defined as:

$$C_{i\leftarrow j,t}^g(J) = \sum_{j=1,i\neq j}^{N} \varphi_{ij,t}^g(J) \quad (8)$$

Moreover, the net total directional connectedness, representing country *i*'s influence on the country *t*, is obtained as the difference between connectedness to others and connectedness from others. This is given as:

$$C_{i,t}^g = C_{i\to j,t}^g(J) - C_{i\leftarrow j,t}^g(J) \quad (9)$$

A positive $C_{i,t}^g$ suggests that country *i* exerts a greater impact on country *t* than vice versa, while a negative value suggests that country *i* is driven by country *t*. The $C_{i,t}^g$ being an aggregated measure, tends to mask central underlying dynamics; hence, there is a need to derive the net pairwise directional connectedness (NPDC), which discloses information about the bilateral transmission process between country *i* and *j*:

$$NPDC_{ij}(J) = \varphi_{ji,t}^g(J) - \varphi_{ij,t}^g(J). \quad (10)$$

If $NPDC_{ij}(J) > 0$ ($NPDC_{ij}(J) < 0$) variable i drives (is driven by) variable j. These results have important implications for property and investment portfolio managers in emerging economies like South Africa, which has huge potential for foreign direct investment inflows.

3.2. Maximal Overlap Discrete Wavelet Transformation (MODWT)

The Maximal Overlap Discrete Wavelet Transformation (MODWT) is the preferred model for confirming the robustness of the findings of the TVP-VAR model. This is because it assists in decomposing the analysis into holding periods, allowing the connectedness to be derived and verified. The advantage of the model is such that it can accommodate any sample size and there is no need for the sample size to be a multiple of 2 or stationary (Percival and Walden, 2000). Furthermore, MODWT does not decimate the coefficients (Cornish et al., 2006). This means that at each transformation point, the number of scales and wavelet coefficients is identical to the sample observations. The wavelet correlation parameters and wavelet covariance at a scale of j between X and Y are obtained by using two formulas of MODWT:

$$\hat{V}_{Xy}^2(T_i) = \frac{1}{M} \sum_{T-l_j}^{N=1} \underline{d}_{j,t}^{(X)} \underline{d}_{j,t}^{(Y)} \quad (11)$$

$$\rho\hat{}_{X,Y}(\tau_j) = \frac{V\hat{}_{X,Y}(\tau_j)}{\sqrt{V\hat{}_{X,X}(\tau_j) V\hat{}_{Y,Y}(\tau_j)}} \quad (12)$$

Where $|\rho\hat{}_{X,Y}(\tau_j)| \leq 1$ indicates that the wavelet correlation is analogous to its Fourier equivalent, the complex coherency (see Gençay et al., 2002). The study incorporates four time scales, follows the time scale of Niu et al. (2023), D1: 2 months, D2: 4 months, D3: 8 months, D4: 16 months. The short-term, medium-term and long-term investment horizons are given by (D1-D2), (D2-D3), and (D3-D4) respectively. The multi-asset time series data were decomposed using MODWT. Thereafter, the wavelet correlations associated with MODWT were generated and compared across various investment horizons.

4. Results

Table 1 presents the average dynamic connectedness results derived from the TVP-VAR framework, showing how shocks are transmitted between EPU and FDI. The Total Connectedness Index (TCI) of 1.15 is relatively low (low degree of interconnectedness), indicating that these variables are largely driven by their own shocks rather than cross-variable spillovers. Nevertheless, the presence of spillovers, though modest, is economically meaningful. FDI exhibits strong internal persistence, with approximately 98% of its forecast error variance explained by its own innovations, while only 1.98% is explained by EPU shocks. In contrast, EPU is even more self-driven, with over 99% of its variance explained by its own shocks. This reflects the fact that policy uncertainty is largely determined by domestic political, institutional, and macroeconomic developments rather than external capital inflows. The net directional connectedness measures indicate that EPU is a net transmitter of shocks, whereas FDI is a net receiver. This asymmetry suggests that changes in policy uncertainty influence foreign investment decisions, but increases or declines in FDI do not materially affect the level of policy uncertainty. The result aligns with theoretical expectations that uncertainty affects investment behavior rather than the reverse.

Table 1. Average dynamic connectedness result

	FDI	EPU	FROM
FDI	98.02	1.98	1.98
EPU	0.33	99.67	0.33
TO	0.33	1.98	2.31
Inc. Own	98.36	101.64	cTCI/TCI
NET	-1.64	1.64	2.31/1.15
NPT	0.00	1.00	

Notes: cTCI is the total connectedness index associated with the bidirectional spillover relationship.

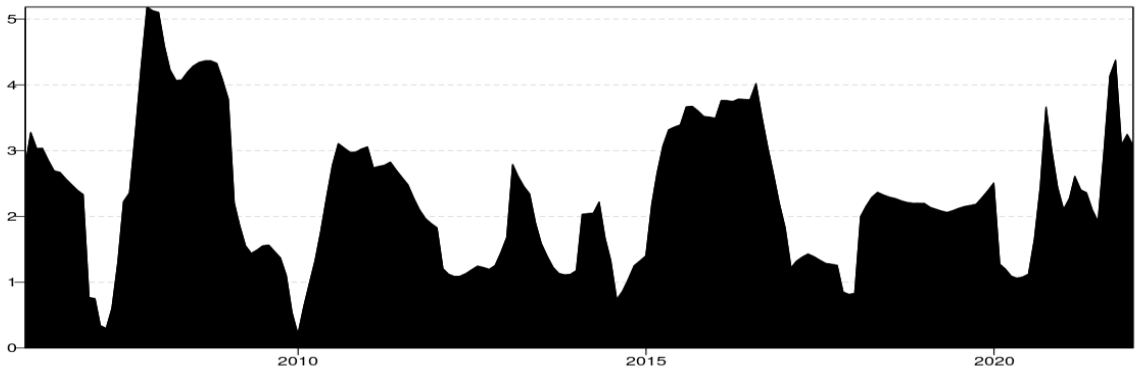


Figure 1. Dynamic connectedness of FDI and EPU

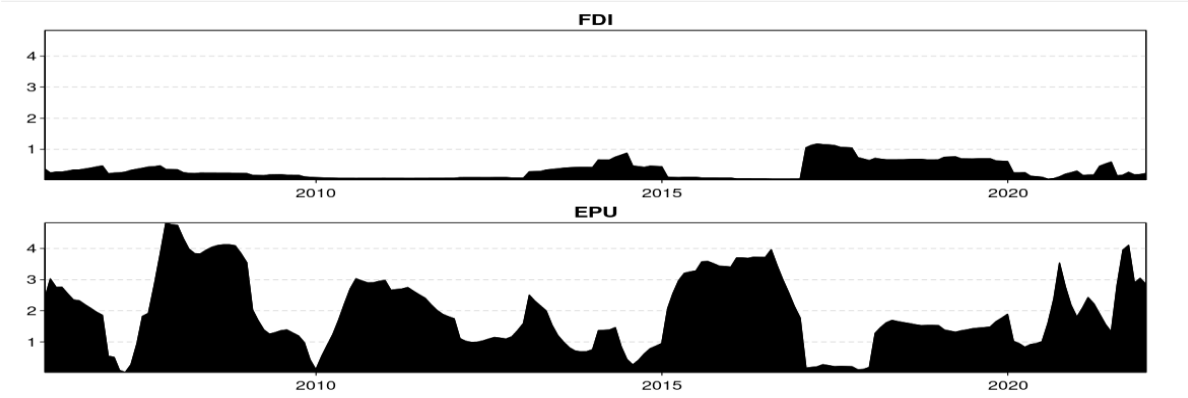


Figure 2. TO connectedness

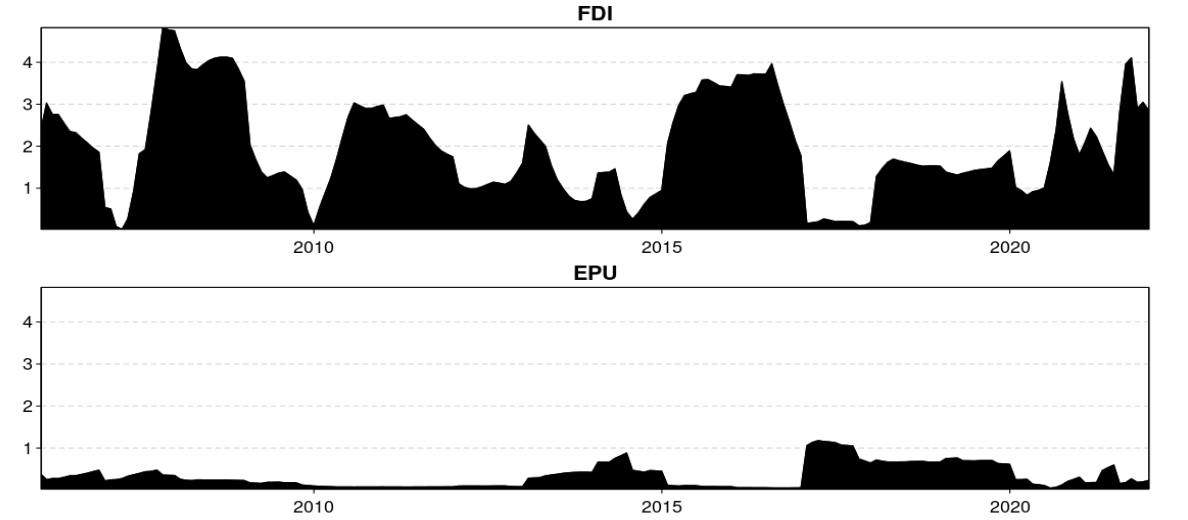


Figure 3. FROM connectedness

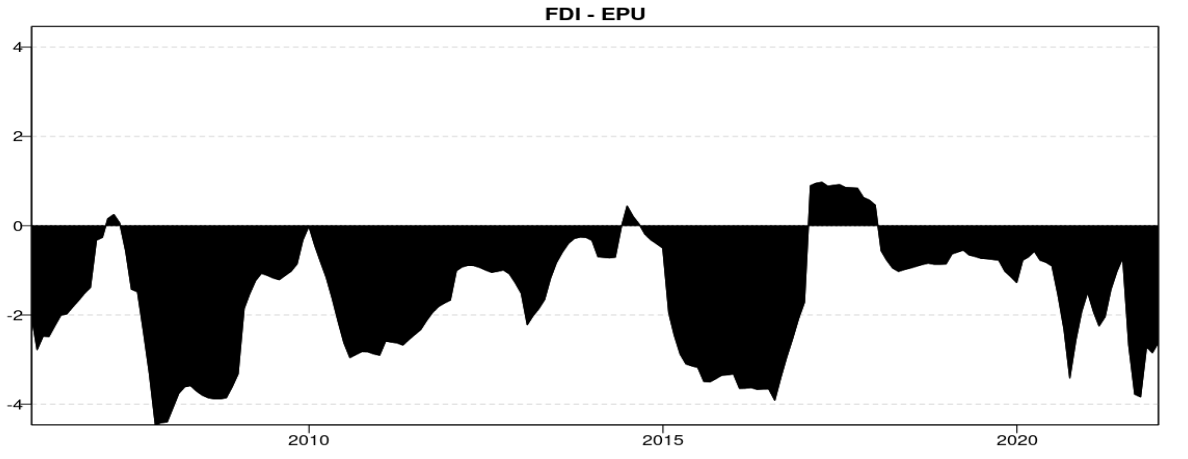


Figure 4. Net pairwise directional connectedness

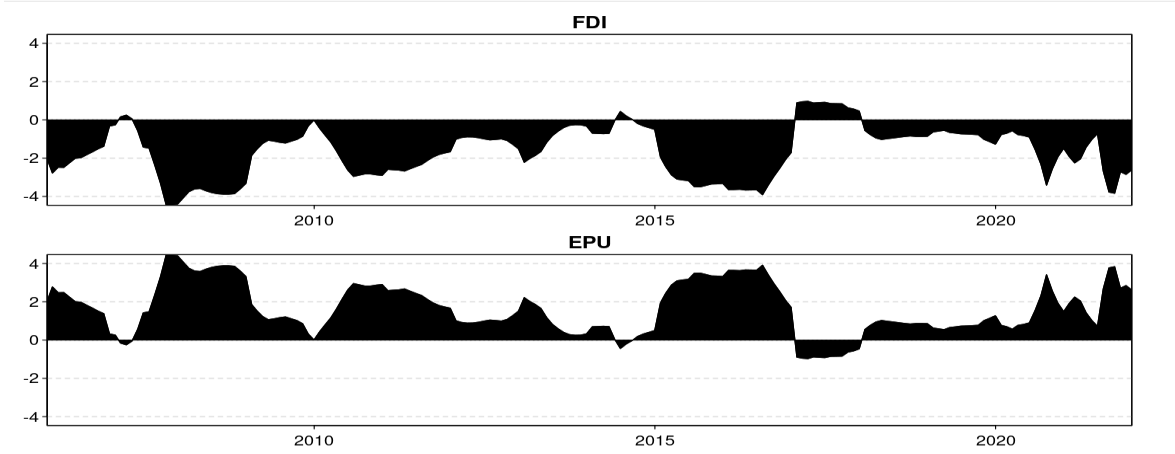


Figure 5. Net total directional connectedness

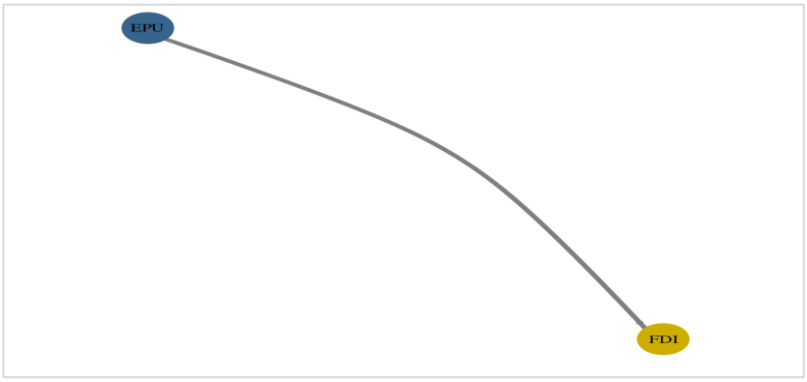


Figure 6. Bidirectional spillover relationship plot between FDI and EPU

Figure 1 illustrates the time-varying evolution of total connectedness between EPU and FDI. The connectedness index fluctuates over the sample period, indicating that the relationship between policy uncertainty and foreign investment is dynamic rather than constant. Periods of elevated connectedness correspond to episodes of heightened economic and political uncertainty, suggesting that spillover effects intensify during times of instability. Conversely, during relatively stable periods, connectedness declines, implying weaker transmission of uncertainty shocks to investment flows. This highlights the importance of accounting for time variation when analyzing uncertainty–investment relationships.

Figure 2 reports the directional connectedness TO others, which measures the extent to which each variable transmits shocks to the rest of the system. The figure clearly shows that EPU consistently transmits more shocks than FDI. This result confirms that EPU acts as a dominant source of volatility, influencing investment behavior through confidence, risk perception, and expectations channels. FDI, by contrast, plays a limited role in transmitting shocks, reinforcing the view that investment inflows are largely reactive rather than causal in shaping uncertainty.

Figure 3 presents the directional connectedness FROM others, capturing the degree to which each variable receives shocks from the system. FDI records higher FROM connectedness values, indicating that it is more exposed to shocks originating from EPU. This suggests that foreign investment inflows are sensitive to changes in the policy environment and respond strongly to uncertainty shocks. EPU, on the other hand, remains relatively insulated from FDI movements, further underscoring the asymmetric nature of the relationship.

Figure 4 depicts the net pairwise directional connectedness between EPU and FDI. The figure shows that net spillovers predominantly run from EPU to FDI across most of the sample period. This persistent direction of influence indicates that increases in policy uncertainty

systematically reduce or destabilize foreign investment inflows, while reductions in uncertainty do not necessarily generate immediate or proportional increases in FDI. The result reflects the irreversible nature of investment decisions and the tendency of investors to delay entry under uncertainty.

Figure 5 summarizes the net total directional connectedness of each variable over time. EPU consistently appears as a net transmitter of shocks, while FDI remains a net receiver. The magnitude of net transmission from EPU increases during periods of heightened uncertainty, highlighting its role as a systemic driver of investment dynamics. This reinforces the conclusion that stabilizing policy uncertainty is crucial for sustaining foreign investment flows.

Figure 6 provides the bidirectional spillover relationship plot between FDI and EPU. The direction and thickness of the arrows indicate that shocks flow predominantly from EPU to FDI, with relatively weak feedback effects. The bidirectional spillover relationship clearly identifies EPU as the dominant force in the relationship between FDI and EPU, emphasizing its systemic importance. This visual evidence complements the numerical results and reinforces the asymmetric and unidirectional nature of uncertainty spillovers.

Collectively, Table 1 and Figures 1–6 demonstrate that EPU is the primary driver of foreign direct investment dynamics, with effects that are time-varying, asymmetric, and predominantly unidirectional. These results justify the use of dynamic and nonlinear modelling approaches for capturing the EPU–FDI relationship.

4.1. Robustness Check

As alluded to previously, the MODWT model is used to confirm the dynamic nature of the relationship between FDI and EPU. Figure 7 provides the wavelet-based correlation estimates between FDI and EPU. The X-axis provides the decomposed investment periods (2, 4, 8, and 16 months), whereas the Y-axis provides the correlation of each associated investment period. It is evident that in the short run (2-4 months), the correlation is stable and below zero. However, during the medium-term (4-8 months) and long-term (8-16 months), the correlation decreases to -0.2 and -0.25, respectively. These findings align with the TVP-VAR framework as it is evident that the connectedness of FDI and EPU is time-varying and dynamic in nature. Collectively, the MODWT confirms the findings of the TVP-VAR framework enhancing the robustness of the findings.

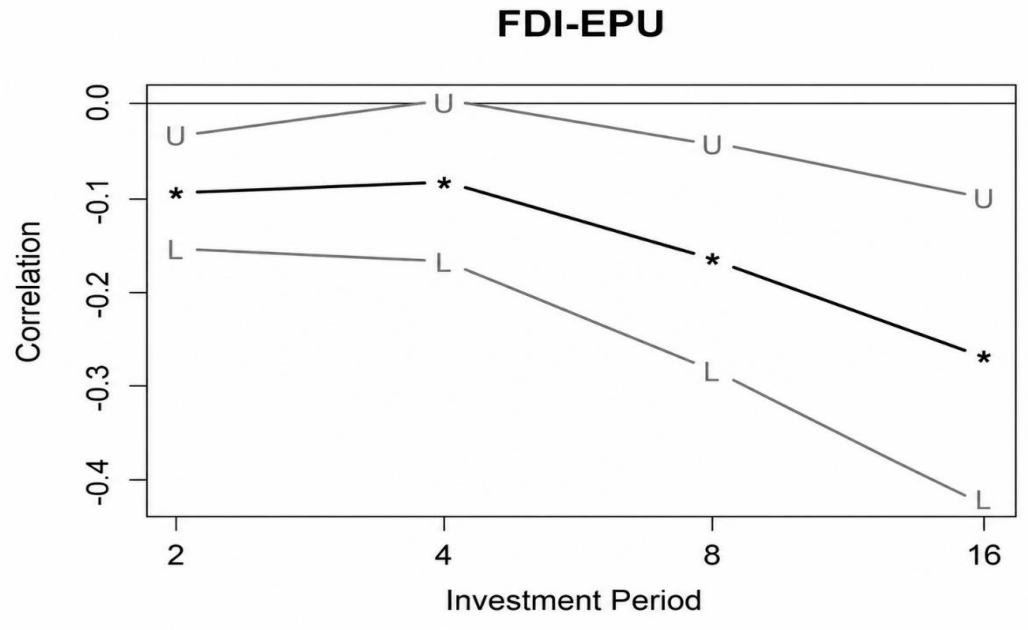


Figure 7. MODWT output

5. Discussion

The empirical results of the TVP-VAR connectedness model support the conclusion that the link between EPU and foreign direct investment (FDI) in South Africa is dynamic, asymmetric, and largely unidirectional. The low TCI indicates that both variables are largely driven by their own shocks, suggesting limited overall spillover effects. Nevertheless, although minimal in amplitude, the nature and duration of spillovers are economically significant, especially in accounting for investment behavior amid uncertainty. The findings consistently demonstrate that EPU is a net transmitter of shocks, but that FDI is a net receiver, thereby indicating that policy uncertainty indeed dominates the foreign investment flow.

This evidence is consistent with theoretical predictions in real options theory (Bloom, 2009; Dixit and Pindyck, 1994) where the uncertainty raises the option value of waiting and hence the decision among investors to delay or reduce irreversible investments (e.g. FDI). The absence of reverse causality from FDI to EPU, also corroborates the view that policy uncertainty is largely exogenous and is influenced by the institutional and macroeconomic factors, and not that it is due to capital arrival (Baker et al., 2016; Pástor and Veronesi, 2013). The time-varying nature of connectedness, as seen in Figures 1–5, indicates that the strength of spillovers varies across different economic conditions. The correlation of periods of increased connectedness with episodes of economic or political instability supports the notion that uncertainty shocks become increasingly salient in crises (Bloom, 2014). This aligns with regime-dependent theories (Ang and Timmermann, 2012; Hamilton, 1989), which emphasize that uncertainty exacerbates impact during turbulent times.

The results of directional connectedness also show that EPU is continuously sending shocks to FDI, so that the effect of policy instability is especially strong upon investor confidence and expectations. Furthermore, the bidirectional spillover relationship plot (Figure 6) also strengthens these results by demonstrating visually that EPU is the dominant factor, when the input and output change (Figure 7). The role of EPU within this spillover system stresses its systemic significance and supports the contention that stabilizing policy environments is essential for foreign investor engagement. These results align with prior evidence that focuses on the institutional quality and policy credibility aspects playing a significant role in determining FDI inflow (Busse and Hefeker, 2007; Henisz, 2000).

MODWT analysis confirms the robustness of these results. A wavelet decomposition indicates that the dependency between EPU and FDI is dynamic at different time horizons. Though short-term correlations are weak, the negative relationship is strongest in the medium and long term. It appears that EPU negatively impacts investment over time, implying the slower nature of investors' reaction to prolonged uncertainty. These results corroborate research emphasising the long-run impact of uncertainty on investment (Julio and Yook, 2012). Importantly, the MODWT findings build on the TVP-VAR findings to show that the EPU–FDI nexus is not only time-varying but also frequency-dependent. This multi-scaled view opens perspective on the effect of uncertainty on investment decisions in different time frames and thus increases the robustness and credibility of the conclusions. In summary, the results add to the literature by illustrating that the relationship between EPU and FDI is more intricate than static models have suggested before. This study uses this lens to capture not only temporal fluctuations but also directional spillovers so that a richer understanding of the role of EPU in investment behavior is achieved in emerging markets.

5.1. Conclusion

This research analyses the relationship between EPU and foreign direct investment in South Africa through the TVP-VAR connectedness framework and MODWT analysis. Thus, the findings of our study show that the connectivity of EPU and FDI is relatively weak; however, the direction and the time-varying nature of the spillovers matter greatly. The results suggest

that EPU is still a persistent net sender of shocks and FDI is a net receiver of shocks, demonstrating a unidirectional relationship with policy uncertainty. In other words, policy uncertainty is important for foreign investment decision-making. Policy uncertainty is not significantly affected by differences in FDI. This time-variant analysis shows that the degree of this correlation becomes deeper during periods of economic and political instability. The MODWT results also confirm that EPU might be considered more influential on FDI when medium- to long-term horizons are included, so it is a significant point to keep in mind when analyzing effects of uncertainty. Together, these two sets of findings suggest that the EPU–FDI relationship is dynamic, asymmetric, and dependent on both time and frequency. Focused on an emerging market characterized by much untested empirical evidence, it fills a critical void in the literature. Unlike static models built on old models, this paper applies a dynamic connectedness approach to examine evolving relationships and directional spillovers. The addition of wavelet analysis for a multi-horizon perspective takes this one step further, achieving a more holistic view of the uncertainty–investment link.

This study has significant implications for policymakers based on how the findings can be used in emerging economies such as South Africa to improve policymaking. The major implication for policymakers is that EPU as a transmitter of shocks emphasizes the importance of coherent, straightforward and transparent, predictable policy instruments. Such reduced policy uncertainty can bolster investor confidence and encourage foreign capital inflow and sustained foreign direct investment. Second, the episodic quality of connectedness indicates that policymakers should take countercyclical and proactive policy adjustments particularly in periods of higher uncertainty. Clear communication of policy intentions and consistent implementation can offset the negative impact of uncertainty on investment. Third, long-term effects of uncertainty observed by MODWT analysis confirm the relevance of the policy to institutional development and credibility of the policy in the long run. Reforms in areas such as the quality and clarity of governance, regulatory standards, and institutional stability will lower uncertainty, thus making South Africa one of the most attractive destinations as a point for investment. Finally, the paper emphasizes the need for policies to make use of dynamic and multi-scale analytical processes in assessing policies. Policymakers may appreciate that the impact of uncertainty is not immediate and may vary across different time horizons, in that the effects can depend on individual policy responses that are tailored to them.

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